Required Capstone Checkpoint 4.1: Advanced Retrieval Implementation

Directions

Download this worksheet and save a copy to your device before you begin working. Use this worksheet to plan, organize, and complete your submission for Required Capstone Checkpoint 4.1. Type your responses directly into the document, save your completed work, and upload the finished file when you are ready to submit.

Completion Checklist

☐ Identification of a meaningful retrieval failure or limitation supported by evaluation evidence
☐ Analysis of the likely cause of the retrieval failure
☐ Selection and justification of an advanced retrieval strategy
☐ Evidence of implementation of the updated retrieval architecture
☐ Evaluation results comparing the updated system against the baseline system
☐ Reflection on the effectiveness of the architectural changes and any remaining limitations

Step 1. Retrieval Failure Identification

Identify a meaningful retrieval limitation in your current retrieval system.

- Describe the failure or limitation.
- Explain how the limitation affects retrieval performance.
- Reference evaluation evidence, test results, queries, or outputs that demonstrate the issue.

Retrieval failure or limitation
Response: Single-pass vector retrieval under-recalled multi-source questions spanning related Wikipedia articles. A required article could share little wording with the query, so it remained absent even when it was explicitly connected to a retrieved article. The answer model then received incomplete evidence.

Evidence supporting the identified limitation
Response:

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Across three runs on the 19-question ground-truth set, the k=7 Markdown baseline achieved 89.9% mean article recall and 78.9% full source coverage. GT16 missed Barack Obama (0.50 recall), GT14 missed Emma Stone (0.75), and GT03 found only one of three required Oscar ceremony articles (0.33).

Step 2. Failure Analysis

Analyze why the retrieval failure occurs. Consider questions such as:

- Does the query require multiple retrieval steps?
- Is relevant information distributed across multiple documents, conversations, or entities?
- Is excessive irrelevant information being retrieved?
- Are important relationships within the data not represented effectively?
- Are there other factors contributing to the failure?

Likely cause(s) of the retrieval failure
Response: The evidence was distributed across related articles, but the baseline indexed only 1,600-character overlapping text chunks. It did not use hyperlinks, shared entities, or categories. Common hub entities also create noise, so unfiltered expansion would retrieve too many weakly related pages.

Why the current retrieval architecture struggles
Response: The baseline performed one vector search and scored chunks independently. It had no path from a strong seed to a related article unless that neighbor also ranked in the top seven semantic matches. Increasing k would add context without targeting the missing relationship.

Step 3. Retrieval Strategy Selection

Select an advanced retrieval strategy designed to address the identified limitation. Examples may include:

- Query rewriting
- Query decomposition

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- Retrieval fusion
- Multi-hop retrieval
- Relationship-aware retrieval
- Graph-based retrieval
- Another advanced retrieval approach

Selected retrieval strategy
Response: Graph-based, relationship-aware retrieval based on the Module 4 Lab 4.2 pattern.

Justification for the selected strategy
Response: The corpus contains about 17,000 in-corpus hyperlinks, and 80% of articles share a category. These signals address the missing-path failure. Lab 4.1 decomposition produced 89.6% or 90.8% mean recall, versus 89.9% for the baseline. Graph expansion was the stronger retrieval change.

Step 4. Updated Retrieval Architecture

Describe the modifications made to your retrieval system.

- Summarize the architectural changes.
- Explain how the updated approach differs from the baseline system.
- Include implementation evidence where appropriate.

Description of architectural changes
Response: I reorganized 2,417 articles into structure-preserving Markdown and JSON sidecars containing titles, sections, infobox fields, table rows, categories, and classified links. A cached WikiGraph represents directed hyperlinks plus shared-entity and shared-category relationships. Vector search returns seven seeds; deterministic one-hop expansion favors rare co-occurrences, drops buckets larger than 50, and adds up to five neighbors (three per seed). Full seed articles and bounded neighbor excerpts form the context. The LLM does not choose graph edges; it synthesizes the answer. In the decomposition experiment, an LLM generated two to four subqueries and optional reasoning instructions.

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Implementation evidence (e.g., screenshots, code snippets, retrieval outputs, notebook excerpts)
Response: Implementation is in retrievers/graph.py and capstone_checkpoint_4_1_graph_retrieval_starter.py. The cached graph contains about 17,000 links, 79,000 entity buckets, and 5,100 category buckets. For GT16, direct and rare shared-entity edges recovered Barack Obama.

Step 5. Evaluation Results

Evaluate the updated system using the evaluation framework developed in previous modules.

- Present evaluation results.
- Compare performance against the baseline system.
- Include representative examples.

Evaluation Metric	Baseline Result	Updated Result
Mean article recall / full source coverage	89.9% / 78.9%	93.9% / 89.5%
Mean answer quality	0.754	0.754 (no change)

Example 1: Query, retrieved information, and result
Response: For GT16, the baseline missed Barack Obama (0.50 article recall). Graph expansion recovered the article through direct-link and shared-entity paths, raising recall to 1.00.

Example 2: Query, retrieved information, and result
Response: For GT14, baseline recall was 0.75 because it missed Emma Stone. The graph recovered her through a direct link and the Academy Award for Best Actress relationship, raising recall to 1.00.

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Example 3 (optional)
Response: GT03 remained unresolved. The required 68th and 74th Academy Awards articles ranked below the five-neighbor cap, so recall remained 0.33.

Step 6. Architectural Impact Analysis

Analyze the impact of the retrieval strategy.

- What improved?
- What remained challenging?
- Did the strategy address the original failure?

Improvements observed
Response: On the development set, mean recall increased from 89.9% to 93.9%, and full source coverage increased from 78.9% to 89.5%. GT14 and GT16 reached complete recall through relationships the vector index had ignored.

Remaining limitations or challenges
Response: Development answer quality was unchanged at 0.754. On validation, graph recall matched the baseline at 97.3%, while quality was 0.885 versus 0.907. Mean development context grew from 207,100 to 260,700 characters. Bounded 12,000-character neighbor excerpts can omit a relevant section, while full articles can exceed the context budget. Graph retrieval did not solve large-table aggregation failures such as GT03, GT10, and GT11. Small test sets and LLM-based answering and judging also add variance.

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Did the strategy address the original failure?
Response: Yes, for the targeted multi-source under-recall: the graph recovered both missing articles. It did not resolve the reasoning and aggregation bottleneck, and the extra context did not improve overall answer quality.

Step 7. Reflection and Next Steps

Reflect on the effectiveness of the architectural redesign.

- What did you learn from implementing this strategy?
- What would you improve next?
- Are there additional retrieval failures that still need to be addressed?

Reflection on the architectural redesign
Response: I learned that article discovery, context selection, and reasoning are separate levers. Rare relationships provide useful signal, while common entities create noise. Finding the right article does not guarantee that the model sees the right section or computes correctly over a table, so recall improved at a cost to context size and precision.

Specific improvements or next steps
Response: Next, I would retrieve query-relevant sections from graph neighbors and route table questions to deterministic filtering, counting, or intersection over sidecar rows. I would verify both changes with repeated matched runs on a larger validation set.

AI Assistance Disclosure

I used Claude and ChatGPT to assist with organizing my code, including refactoring my project to extract common checkpoint logic (see utils/ragas_eval), as well as creating a graph database utilizing categories, article links. I have also accepted AI suggestion to prune categories that were too generic (over parameterized number of connections).

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In addition, I used ChatGPT to organize my observations and refine the wording of my written analysis. I reviewed the prompts, experimental runs, interpretations, and final conclusions and made all final decisions regarding final outcomes and evidence to be taken into account for this report.

End of Required Capstone Checkpoint 4.1

Return to the assignment page in the learning platform to upload and submit your completed capstone checkpoint and complete any remaining items in this module.

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